

# Module 5: Markdown In, HTML Out

Markdown & HTML Crash Course — Panaversity Agent Factory

Prepared by: Imran Hassan

**50**

Total MCQs

**14**

Concepts

**0**

Live Score

0 / 50 answered



## Concept 1 & 2: Why Agents Need Structure + The Asymmetry

### SCENARIO

Maria asks an AI agent: "Can you build me a nice website for my bakery?" The AI returns something that doesn't match her vision at all — wrong colors, wrong sections, wrong tone.

1 According to Concept 1 of Module 5, why did the AI produce a mismatched result?

- A** The AI model was too old and couldn't understand the request.
- B** The unstructured request forced the agent to infer intent through guesswork, making constraints invisible.
- C** The AI needed a paid subscription to build websites.
- D** Maria should have used a different programming language in her prompt.

### SCENARIO

*A project manager writes a long paragraph describing his requirements to an AI agent. A colleague suggests he rewrite it using headings and bullet points instead.*

2 What is the key benefit of using headings and bullets according to Module 5?

- A They make the prompt look more professional to impress the AI.
- B They make constraints explicit and checkable, so the agent doesn't need to guess intent.
- C They reduce the cost of running the AI tool.
- D They allow the AI to respond in multiple languages.

#### SCENARIO

*Ahmed is writing a prompt to an AI. His most critical requirement — "the deadline is Friday, no exceptions" — is buried inside a long paragraph. His mentor tells him to restructure the prompt.*

3 What rule from Module 5 Concept 1 applies here?

- A Critical requirements should be bolded inside paragraphs for emphasis.
- B Anything an agent must not get wrong belongs under its own heading or bullet — never inside a sentence.
- C Long paragraphs are preferred by AI agents for better understanding.
- D Only technical requirements need to be in bullets; deadlines can stay in text.

#### SCENARIO

*A team uses Markdown to write specs sent to an AI agent and asks the agent to return a polished, shareable report. They ask: "Why Markdown for input and HTML for output?"*

4 What is the "asymmetry" concept in Module 5 that explains this practice?

- A Markdown is for designers; HTML is for developers — they have different roles.
- B Markdown going in removes ambiguity for machines; HTML coming out provides rich, readable, shareable output for humans.
- C HTML is for input because it is more powerful than Markdown.
- D Both formats should be used randomly depending on mood.

#### SCENARIO

*Two AI agents are passing task instructions to each other in a pipeline. A developer asks: "Should I configure them to exchange data in HTML or Markdown?"*

5 According to Module 5 Concept 2, what format should Agent-to-Agent communication use?

- A HTML — it is richer and more expressive.
- B Markdown — it stays compact and efficient for machine processing.
- C Plain text with no formatting at all.
- D XML — it is the most structured format for data exchange.

#### SCENARIO

*A student is unsure whether to request a Markdown or HTML response from an AI agent. Her instructor tells her to ask one key question to decide.*

6 What is the deciding question from Module 5 Concept 2 that determines format choice?

A "How long is the content?"

B "Who reads this last?"

C "Which AI model am I using?"

D "How technical is the content?"



## Concepts 3–5: Headings, Lists, and Code Blocks

### SCENARIO

*A developer writes a Markdown spec with the following heading structure: ``# Title``, then immediately ``### Section``, skipping ``##`` entirely. His reviewer flags this as incorrect.*

7 Which heading rule from Module 5 Concept 3 was violated?

A You should use multiple ``#`` symbols on the same document title.

B Heading levels should never be skipped — ``#`` must be followed by ``##``, not ``###``.

C All headings must be written in uppercase letters.

D You cannot use more than two heading levels in one document.

**SCENARIO**

*An analyst writes a spec section titled "## Budget" with nothing else under it. Her colleague suggests changing it to "## Budget: PKR 50,000 hard ceiling."*

8 Why is the colleague's version better according to Module 5 Concept 3?

- A** Longer headings are always better for SEO purposes.
- B** Headings should be declarative — stating the actual constraint, not just a label — so the AI gets the value directly.
- C** Numbers in headings make them process faster in AI tools.
- D** The original "Budget" heading would confuse the file system.

**SCENARIO**

*A product manager lists the steps to deploy a new feature: "1. Run tests, 2. Merge to main, 3. Deploy to staging, 4. Get approval, 5. Deploy to production." His teammate says using a bullet list (•) instead of numbers would be wrong here.*

9 Why is the numbered list the correct choice according to Module 5 Concept 4?

- A** Numbers look more professional in technical documents.
- B** Numbers signal a sequence where order is critical — the AI treats them as steps that must be followed in order.
- C** Bullet lists are only for personal notes, not professional prompts.
- D** AI agents cannot read bullet lists, only numbered lists.

**SCENARIO**

*A designer lists her color preferences, fonts, and image style in a spec: "- Blue tones, - Sans-serif font, - Minimalist photography." She is debating whether to use bullets or numbers.*

10 According to Module 5 Concept 4, which list type is correct here and why?

- A** Numbered list — all design requirements must be followed in a specific order.
- B** Bullet list — these are a set of preferences where order doesn't matter, signaling they are independent requirements.
- C** Neither — design specs should always be written in paragraphs.
- D** Both types are interchangeable — it makes no difference which one you choose.

**SCENARIO**

*A developer pastes an error message into a prompt without any formatting. The AI misreads parts of it as instructions. His colleague says he should have wrapped the error message in triple backticks.*

11 What is the purpose of triple backtick code blocks (```) in AI prompts according to Module 5 Concept 5?

- A** They tell the AI to execute the wrapped code immediately.
- B** They wrap text the AI should analyze or reference — not execute — preventing misinterpretation as instructions.
- C** They are only useful for Python code, not error messages.
- D** They make the text bold in the AI's response.

**SCENARIO**

*A data analyst wants the AI to generate a CSV report in a very specific format. Instead of describing the format in three paragraphs, her senior says "just show one example row."*

12 Which Module 5 key rule does this advice reflect?

- A** Always describe formats in detail so the AI has full context.
- B** "One example row beats three paragraphs of format description" — show, don't tell.
- C** CSV formats do not need to be specified — AI generates them automatically.
- D** Format descriptions should always be in numbered lists, not code blocks.

**SCENARIO**

*A content editor wants to share a quote from an article with an AI and ask it to analyze the tone — not rewrite it. She wraps the quote in triple backticks before pasting it.*

13 What specific benefit does wrapping the quote in triple backticks provide according to Module 5?

- A** It makes the quote appear in a larger font for the AI to see better.
- B** It quotes external text without ambiguity, signaling the AI should analyze it, not treat it as part of the instruction.
- C** It compresses the text to save tokens in the AI model.
- D** It forces the AI to respond only in plain text.



## Concepts 6 & 7: Links, Images, and the Spec Skeleton

### SCENARIO

*A student describes her university's fee refund policy in her own words inside a prompt. Her instructor says she should just include the link to the actual fee policy page instead.*

14 Why is including the link better according to Module 5 Concept 6?

- A** Links look more impressive in AI prompts.
- B** Links enable agents to read actual source documents directly, ensuring accuracy over paraphrasing.
- C** AI agents cannot read text — they only process URLs.
- D** Paraphrasing is always more accurate than linking to sources.

### SCENARIO

*A team is building a spec for an AI-generated report. They have a Goal section, Context, and Requirements — but their reviewer says the spec is still incomplete and gives it a 6/10.*

15 According to the Specification Skeleton in Concept 7, which three sections are most likely missing?

- A** Introduction, Body, and Conclusion
- B** Hard Constraints, Out of Scope, and Expected Output
- C** Abstract, Methodology, and References



**D** Team Name, Budget Summary, and Timeline**SCENARIO**

*A spec includes: "Build a dashboard with charts, filters, export to PDF, user authentication, mobile app version, and dark mode." The AI builds all of it — but the client only wanted the basic dashboard.*

16 Which Concept 7 section was missing from the spec that caused this over-delivery?

**A** The Goal section — it was too vague.

**B** The "Out of Scope" section — explicitly stating what NOT to build prevents over-delivery.

**C** The Context section — more background would have helped.

**D** The Requirements section — it needed more bullet points.

**SCENARIO**

*Before asking an AI to build an app, a developer rates her spec and gives it a 7/10 for clarity. Her mentor says she should not proceed yet.*

17 What does Module 5 Concept 7 say about the minimum score before execution?

**A** A 5/10 is sufficient — perfection is the enemy of progress.

**B** Grade the spec and iterate until it reaches 9+ out of 10 before execution.

**C** Grading specs is a waste of time — just build and iterate.

**D** A 7/10 is fine — nothing will ever be perfect.

#### SCENARIO

*A startup founder wrote: "Requirements: The product should be fast, look good, and work well." Her technical advisor says these are not proper checkable bullets.*

**18** Which example from Module 5 best illustrates a proper checkable bullet?

**A** "The app should be user-friendly and responsive."

**B** "Page loads in under 3 seconds on a 3G connection."

**C** "The design should be modern and attractive."

**D** "The product should work well on all devices."

#### SCENARIO

*A spec's "Expected Output" section says only: "A nice report." An engineer suggests adding a fenced example or a specific format description instead.*

**19** Why is the engineer's suggestion better according to Module 5 Concept 7?

**A** AI agents prefer technical jargon over simple language.

**B** A fenced example or format description gives the AI a concrete, unambiguous target to match exactly.

**C** The AI will always produce a "nice report" regardless of instructions.

- D** Expected Output sections should be removed to keep specs short.

**SCENARIO**

*A developer spent 3 hours fixing an app the AI built from a bad spec. His manager says: "You should have fixed the spec first — it would have taken one minute."*

**20** Which key rule from Module 5 does the manager's statement reflect?

- A** "The out-of-scope section kills over-delivery."
- B** "Grade before you build — fixing a spec costs one minute; fixing what the agent built from a bad spec costs an afternoon."
- C** "Always ask for HTML output regardless of the task."
- D** "One example row beats three paragraphs of format description."

**Concepts 8 & 9: Why HTML + How to Prompt for It****SCENARIO**

*An AI agent returns a 200-line Markdown report to a busy executive. The executive skims the first 20 lines and stops reading. The team says they should have requested HTML output instead.*

**21** According to Module 5 Concept 8, why does long Markdown fail as a deliverable?

- A** Markdown files are too large to email.
- B** Long Markdown (past ~100 lines) becomes unread — HTML provides density, navigability, and shareability.

**C** Executives don't understand Markdown formatting.

**D** Markdown cannot contain data tables or charts.

#### SCENARIO

*A team wants to share an AI-generated analysis with 50 stakeholders. Some want to click interactive filters; others want to share a link. They ask if HTML output supports this.*

22 Which benefits of HTML output does Module 5 Concept 8 highlight for this scenario?

**A** HTML can only display static text — no interactivity possible.

**B** HTML provides shareability as a link, interactivity (sliders, toggles), and navigability — all at once.

**C** HTML requires a paid web server for all sharing.

**D** HTML is only suitable for developers, not general stakeholders.

#### SCENARIO

*A user asks an AI: "Give me an HTML report with `*

*, and `` tags." Her colleague says she is prompting for HTML the wrong way.*

23 What is the correct way to prompt for HTML according to Module 5 Concept 9?

**A** Always specify exact HTML tags so the AI knows exactly what to write.

**B** *Never describe HTML tags. Instead, specify who reads it, what it contains, interactivity needs, and reading context.*

**C** *Write CSS styles in the prompt so the AI applies them directly.*

**D** *Use JavaScript code snippets in the prompt to define the layout.*

#### SCENARIO

*A founder wants an HTML investor report. She writes the prompt: "For my two co-founders on laptops, reading once: headline numbers, monthly revenue chart, top 5 growers table, and recommendations section."*

24 *According to Module 5 Concept 9, which four elements does this strong prompt correctly specify?*

**A** *Font size, color scheme, animation speed, and browser compatibility*

**B** *Who reads it, what it contains, interactivity needs, and reading context*

**C** *HTML version, CSS framework, JavaScript library, and hosting platform*

**D** *Word count, paragraph count, image count, and link count*

#### SCENARIO

*After receiving an HTML report from an AI, a user tells the AI: "The heading is too quiet — make it bolder and add a button to download the data." This refinement loop is described in Module 5.*

25 *What does Module 5 call this capability of making visual adjustments through feedback?*

**A** *Code debugging loop*

**B** *"Copy back" loop — enabling iterative visual adjustments by giving targeted feedback on the HTML output*

**C** *Markdown conversion cycle*

**D** *API refresh pattern*

## Concepts 10 & 11: Five HTML Patterns + Social Media

### SCENARIO

*A product team is comparing three pricing models side by side to decide which to launch. They ask an AI for an HTML output to help visualize the comparison clearly.*

26 Which of the Five HTML Patterns from Module 5 Concept 10 best fits this use case?

**A** *Throwaway Editors — drag-and-drop prioritization cards*

**B** *Plans/Explorations — comparison grids of options side by side*

**C** *Design Prototypes — live color sliders*

**D** *Code Review — color-coded diffs*

### SCENARIO

*A developer wants to review AI-suggested changes to her codebase. She needs to see what was added, removed, and modified — each clearly color-coded with explanations alongside.*

27 Which HTML pattern from Module 5 Concept 10 is designed for this use case?

**A** Explainers/Reports — one-page summary with diagrams

**B** Code Review — color-coded diffs with annotated explanations

**C** Plans/Explorations — comparison grids

**D** Throwaway Editors — drag-and-drop ranking cards

SCENARIO

*A UX designer wants to try different color palettes for a brand in real-time, adjusting hues with sliders and immediately seeing how the design changes. She asks an AI for an interactive HTML tool.*

28 Which of the Five HTML Patterns from Concept 10 matches this need?

**A** Explainers/Reports

**B** Design Prototypes — live sliders/swatches to test design choices interactively

**C** Code Review

**D** Plans/Explorations

## SCENARIO

*A product manager has a backlog of 20 features to prioritize. She wants to drag them around on screen to rank them, then paste the final ranking back to the AI for the next step.*

29 Which of the Five HTML Patterns is this from Module 5?

**A** Explainers/Reports

**B** Throwaway Editors — drag-and-drop cards to prioritize/sort; copy result back as text

**C** Code Review

**D** Design Prototypes

## SCENARIO

*A social media manager uses an AI to write a LinkedIn post with bold headers and bullet points in Markdown. When she posts it, all the formatting disappears and it looks like a wall of text.*

30 Why did this happen according to Module 5 Concept 11?

**A** LinkedIn has a bug that removes all text formatting.

**B** Social media platforms strip Markdown formatting — the actual post must be plain text.

**C** The AI wrote the post in the wrong language.



**D** *Bold and bullets only work on Twitter/X, not LinkedIn.*

**SCENARIO**

*A marketer wants her shared article link to show a compelling preview card on social media — with a custom title, description, and thumbnail. She asks the AI how to control this.*

31 *According to Module 5 Concept 11, what controls how link previews render on social platforms?*

**A** *The color scheme of the website*

**B** *Open Graph meta tags in the HTML — these shape how link preview cards render on social platforms*

**C** *The number of images on the page*

**D** *The domain name of the website determines preview appearance*

**SCENARIO**

*A content creator wants to post a visually rich, formatted graphic on Instagram. She asks an AI to help. Her mentor says to build it as HTML, then save it as a PNG image before posting.*

32 *What does Module 5 say about using HTML for social media image creation?*

**A** *HTML cannot be converted to PNG — use Photoshop instead.*

- B** Build the designed image as HTML, save it as a PNG, then post the image — this bypasses platform formatting restrictions.
- C** Social media platforms now support HTML posts directly.
- D** Plain text is the only effective format for social media posts.



### Concepts 12 & 13: Publishing + Document Formats

#### SCENARIO

A student wants to publish her AI-generated HTML report online immediately with zero setup. She just wants a shareable link right away.

33 Which publishing option from Module 5 Concept 12 is the easiest with zero setup?

- A** GitHub Pages — free permanent web address
- B** Claude.ai publish — easiest option with zero setup; link lives on Claude's servers
- C** Netlify — requires account and deployment setup
- D** GitHub Gist — requires Git knowledge

#### SCENARIO

A developer wants to host an HTML file permanently at a custom free web address, with full version history and the ability to edit the file at any time. He

*is comparing GitHub Gist and GitHub Pages.*

34 What does Module 5 say about the difference between GitHub Gist and GitHub Pages?

- A** They are identical — both provide a permanent free web address.
- B** GitHub Gist lets you own the file and edit it with version history; GitHub Pages provides a permanent free web address (requires setup).
- C** GitHub Gist requires a paid plan; GitHub Pages is free.
- D** GitHub Pages only hosts JavaScript files, not HTML.

**SCENARIO**

*A non-technical freelancer has an HTML file and wants to deploy it permanently for free online right now — without using Git, command lines, or any developer tools.*

35 Which publishing option from Module 5 Concept 12 is best for her?

- A** GitHub Pages — best for non-technical users
- B** Netlify drag-and-drop — simplest permanent hosting; free account, no technical setup needed
- C** GitHub Gist — requires the most technical knowledge
- D** AWS S3 — the industry standard for non-technical users

**SCENARIO**

*An HR manager receives an AI-generated employee evaluation and needs to edit comments, add annotations, and merge it into the company's existing HR documentation system in Microsoft Word.*

36 According to Module 5 Concept 13, which output format should be requested from the AI?

**A** PDF — best for documents that need editing

**B** Word (.docx) — designed for editing, adding comments, and merging into existing documents

**C** CSV — best for human-readable text documents

**D** HTML — all documents should be in HTML format

**SCENARIO**

*A lawyer needs an AI-generated contract that she will sign, print, and file permanently — it must never be edited after creation.*

37 Which output format from Module 5 Concept 13 is designed for sign, print, and permanent file use cases?

**A** Word (.docx)

**B** PDF — designed for signing, printing, and permanent filing

**C** CSV

**D** *Slides (.pptx)*

**SCENARIO**

*A data team generates an AI report with 500 rows of sales data. They plan to feed this data into another analytics tool and need the format to be machine-readable for easy import.*

38 Which format from Module 5 Concept 13 is best for feeding data to another tool or AI?

**A** *PDF*

**B** *Slides (.pptx)*

**C** *CSV — designed to feed data to another tool or AI*

**D** *Word (.docx)*

**SCENARIO**

*A consultant writes a detailed research report in Markdown. Different stakeholders need: a PDF for the board, a .docx for the legal team to edit, and a .pptx for the presentation meeting.*

39 What is the "one-time rule" from Module 5 Concept 13 that applies here?

**A** *Write the content three times — once for each audience format.*

**B** *Write content once in structured form (Markdown for text, CSV for data), then export to whatever format each recipient needs.*

**C** Only use one format for all stakeholders to avoid confusion.

**D** Always deliver everything in PDF — it is universally compatible.

## Concept 14 & The Spec-to-HTML Pipeline

### SCENARIO

A user opens Claude.ai, pastes some notes into the chat, and asks the AI to build a formatted report. An artifact (the report) appears alongside the conversation.

40 Which of the Three Motions (Tools) from Module 5 Concept 14 is this user using?

**A** Terminal — AI reads files directly from folders

**B** Chat — paste notes into the AI; it builds artifacts beside the conversation

**C** Desktop — visual app that proposes a plan before touching files

**D** API — sends requests programmatically

### SCENARIO

A developer runs Claude Code (Terminal mode) on her project. Instead of pasting file contents into a chat, the AI directly reads the source files from her project folders.

41 Which of the Three Motions is this, and what is its key characteristic?

**A** Chat — best for pasting notes manually

**B** Terminal — AI reads files directly from folders on your computer without manual copy-paste

**C** Desktop — proposes a plan before touching anything

**D** Cloud — AI syncs with remote servers automatically

SCENARIO

*A non-developer uses Cowork (an AI desktop app). Before making any file changes, the AI shows her a visual plan of what it intends to do and waits for her approval.*

42 Which of the Three Motions is this and what makes it distinct from Terminal mode?

**A** Chat — it pastes content into a chat window

**B** Desktop — visual app that proposes a plan before touching files, unlike Terminal which acts more directly

**C** Terminal — same as Claude Code

**D** API — sends programmatic requests to the AI server

SCENARIO

*A team learned prompting skills using Claude. They now switch to ChatGPT for a project and wonder if they need to relearn a completely different prompting*

**approach.**

43 What does Module 5 Concept 14 say about prompting across different tools?

- A** Each AI tool requires a completely different prompting language and approach.
- B** The same prompts and patterns work across all three motions — only the input method changes, not the skill.
- C** ChatGPT only accepts HTML prompts, unlike Claude which uses Markdown.
- D** Terminal mode tools don't support Markdown specifications.

**SCENARIO**

**A trainer summarizes the Spec-to-HTML pipeline to students. She describes: messy notes → Markdown spec → grade to 9+ → request HTML with audience details → refine with feedback → publish URL.**

44 In this pipeline, what happens AFTER grading the spec to 9+ according to Module 5?

- A** Immediately publish the spec as-is to a URL.
- B** Ask for an HTML artifact, specifying audience and specific content details, then refine with targeted feedback.
- C** Convert the spec to CSV format for data processing.
- D** Send the spec directly to the client without any further processing.



## SCENARIO

*At the end of Module 5, an instructor asks: "What is the one sentence you should remember from this entire course?"*

45 *What is Module 5's core one-sentence takeaway?*

- A** *"Always use HTML for both input and output — it is the most powerful format."*
- B** *"Write Markdown precise enough for a machine; demand HTML rich enough for a human."*
- C** *"AI tools work best when given unstructured, conversational input."*
- D** *"Use plain text for everything and let the AI decide the format."*

 **Mixed Concept Review — Q46 to Q50**

## SCENARIO

*A spec writer is unsure whether to use a bullet or a numbered list for a list of "things to include" in a report: cover page, table of contents, executive summary, and appendix.*

46 *Based on Module 5 Concept 4, which list type should she use and why?*

- A** *Numbered list — all document sections must appear in a specific order.*
- B** *Bullet list — these are a set of required elements where the exact order is not a hard constraint.*

**C** *Either — they are interchangeable in all situations.*

**D** *Numbered list — numbers always look more professional in prompts.*

**SCENARIO**

*A team's Specification Skeleton says "Goal: Build a mobile app." Their reviewer says this Goal section is insufficient and rates the spec 4/10.*

47 *According to Module 5 Concept 7, what should a proper Goal section contain?*

**A** *A one-word description of the product type*

**B** *One paragraph explaining what is being built and who it serves*

**C** *A list of all the technologies to be used*

**D** *The name of the developer team and their experience*

**SCENARIO**

*A professor asks students to write one habit they will build from Module 5. One student writes: "I will always ask the AI to grade my spec before it builds anything."*

48 *Which Module 5 practical takeaway directly matches this habit?*

**A** *"Keep one picture — the 'who reads this last?' decision tree."*

**B** "One habit to build: Always ask agents to grade your spec before building anything."

**C** "Always use numbered lists instead of bullets for all requirements."

**D** "Write specs in HTML, not Markdown, for better AI understanding."

#### SCENARIO

Module 5 describes three projects that build real deliverables. Project 1 is "messy notes → designed page → published URL." Project 3 turns a tool into a full-screen slide deck exported as a real PowerPoint.

49 What does Project 2 "Spin the Wheel" demonstrate in Module 5?

**A** How to build a website from scratch using HTML alone

**B** Building a tool you operate yourself, then handing its text result back to the AI as input for the next step

**C** How to publish an HTML file to GitHub Pages automatically

**D** How to convert a PDF to a Markdown specification

#### SCENARIO

At the end of Module 5, a student summarizes the asymmetry rule: "Markdown for input, HTML for output." She is asked to explain this using the 'who reads last' decision tree for three cases: a person opening a browser, an AI reading next, and a social media feed.

50 What is the correct format mapping for all three cases according to Module 5?

**A** Person in browser → Markdown; AI reading next → HTML; Social feed → CSV

**B** Person in browser → HTML; AI reading next → Markdown; Social feed → Plain text

**C** All three cases → HTML, since HTML is universally best

**D** Person in browser → PDF; AI reading next → CSV; Social feed → Markdown

Submit Quiz

Reset

---

**Module 5 — Markdown & HTML Crash Course | Panaversity Agent Factory | Prepared by:**  
*Imran Hassan*